

SIMILARITY TRANSFORMATION

Computer Vision with Hüseyin Özdemir

Similarity transformation is a combination of uniform (isotropic) scale (resize), rotation and translation

Similarity means the transformation preserves shape

Isotropic scale, rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise rotation angle
 α is scale factor

Isotropic scale, rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise rotation angle
 α is scale factor

Output coordinate vector Scale matrix Rotation matrix Input coordinate vector Translation vector

The diagram illustrates the transformation equation for isotropic scale, rotation, and translation in Cartesian coordinates. The equation is:
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$
 Below the equation, five labels are connected to their respective components by arrows: 'Output coordinate vector' points to the left vector, 'Scale matrix' points to the first matrix, 'Rotation matrix' points to the second matrix, 'Input coordinate vector' points to the right vector, and 'Translation vector' points to the final vector. To the right of the equation, a text block states: ' θ is clockwise rotation angle' and ' α is scale factor'.

Isotropic scale, rotation and translation in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

θ is clockwise rotation angle
 α is scale factor

Isotropic scale, rotation and translation in homogeneous coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \cos \theta & -\alpha \sin \theta & b_0 \\ \alpha \sin \theta & \alpha \cos \theta & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \cos \theta & -\alpha \sin \theta & b_0 \\ \alpha \sin \theta & \alpha \cos \theta & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Similarity transformation is controlled with 4 parameters

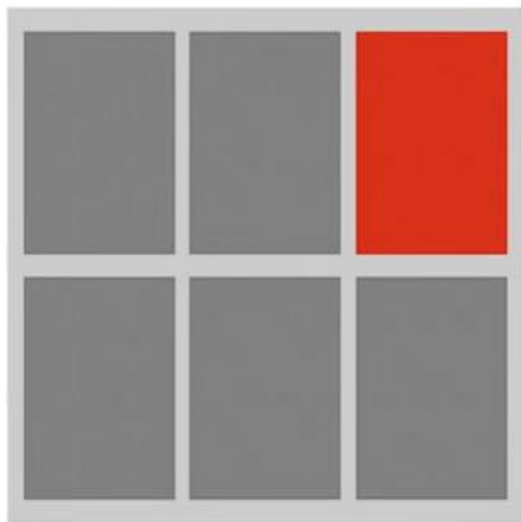
4 Degrees of Freedom (DoF)

α , θ , b_0 and b_1

Similarity Transformation preserves lines, angles
and ratio of lengths over each line

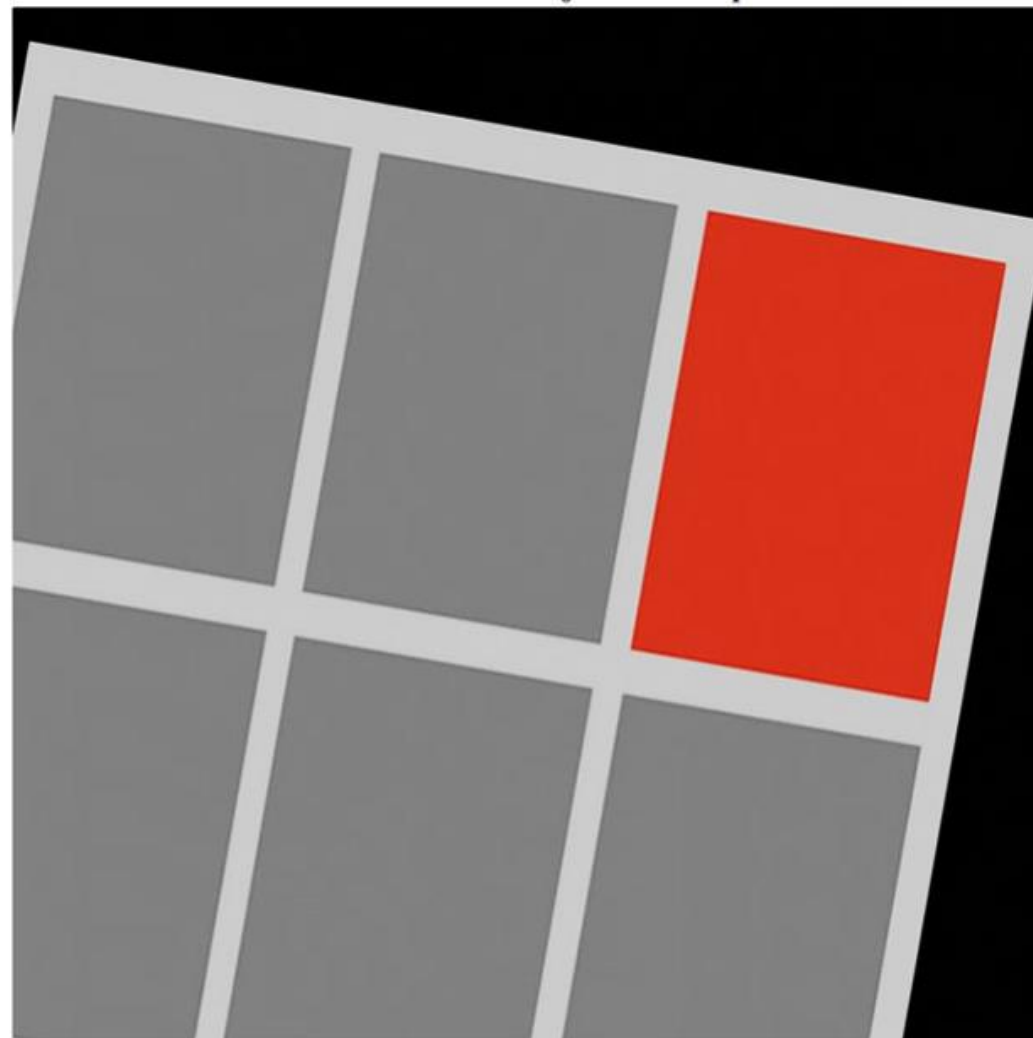
Similarity Transformation may not preserve lengths

Parallel lines remain parallel



Original Image

$$\theta = 10^\circ, \alpha = 2, b_0 = 30, b_1 = 60$$



Rotated, Scaled and Translated Image

Reference

- https://www.youtube.com/watch?v=1-1QEsCPo4g&list=PLcEc12xiYZ7r24dJWP4GYwwXUgGS-wl4_&index=6